PROJECT DESIGN PHASE – II

DATA FLOW DIAGRAMS & USER STORIES

Project Name	ServiceNow-Based Medical Diagnostic Center Management System
Team ID	
Date	25 September 2026
Maximum Marks	4 Marks

1. Data Flow Diagrams

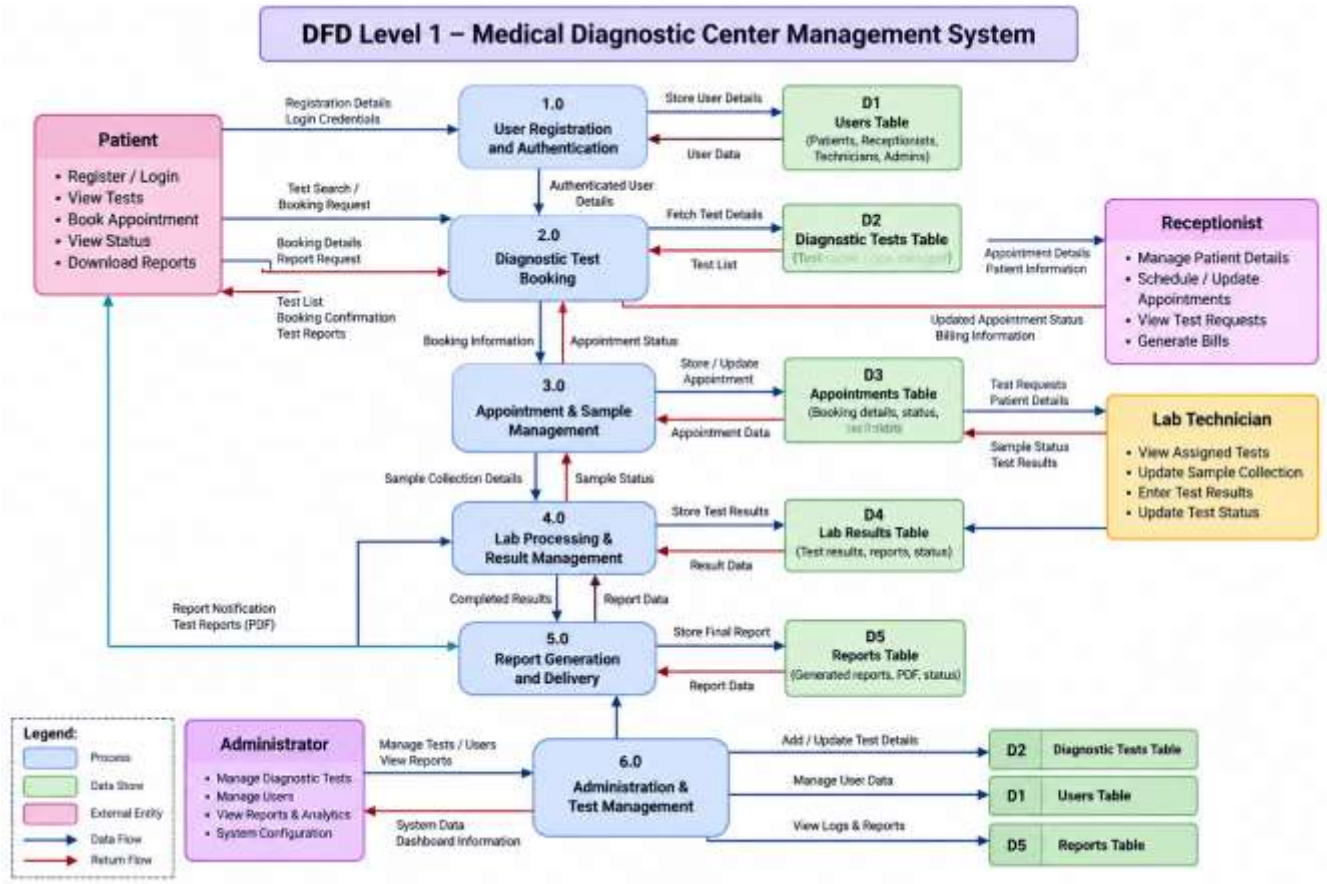
A Data Flow Diagram (DFD) visually represents how information enters, moves through, is processed by, and is stored in a system. The following diagrams are adapted to the Medical Diagnostic Center Management System built on ServiceNow.

1.1 DFD Level 0 – Context Diagram



The system interacts with Patients, Receptionists, Lab Technicians, and Administrators. Patients submit registration and booking information and receive confirmations and reports. Staff process bookings, samples, and results. Administrators maintain the test catalog and system records.

1.2 DFD Level 1 – Detailed Process Diagram



1. Patient Registration
    1. Patient enters details.
    2. System validates and saves patient information.
    3. Registration is confirmed.
2. Diagnostic Test Booking
    1. Patient selects a test and time slot.
    2. System checks availability.
    3. Booking is saved and confirmed.
3. Sample Processing
    1. Lab technician collects samples.
    2. Samples are tested.
    3. Results are stored in the system.

4. Report Generation

1. System retrieves test results.
2. Technician verifies results.
3. Report is generated and shared with the patient.

5. Administration

1. Admin manages patients and tests.
2. Admin manages bookings and reports.
3. System displays updated information on the dashboard.

2. User Stories

The following user stories use the template's columns: User Type, Functional Requirement (Epic), User Story Number, User Story / Task, Acceptance Criteria, Priority, and Release.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Patient	Registration / Login	USN-1	As a patient, I can register or sign in to the ServiceNow portal so that I can access diagnostic services.	I can securely access my patient dashboard after valid sign-in.	High	Sprint-1
Patient	Diagnostic Test Catalog	USN-2	As a patient, I can view available diagnostic tests, descriptions, prices, and durations so that I can choose a required test.	The catalog displays active tests with their details and current availability.	High	Sprint-1
Patient	Appointment Booking	USN-3	As a patient, I can select a diagnostic test, date, and time slot so that I can book an appointment.	A valid booking is saved and a confirmation is shown; the booking is visible in appointment records.	High	Sprint-1
Patient	Appointment Tracking	USN-4	As a patient, I can view my appointments and their status so that I can track my booking.	Only the signed-in patient's appointments are shown with current status.	High	Sprint-1
Receptionist	Patient and	USN-5	As a receptionist, I	Authorized staff	High	Sprint-

	Appointment Management		can view and update patient appointment details so that visits are organized.	can view bookings and update permitted appointment fields.		2
Lab Technician	Sample Processing	USN-6	As a lab technician, I can update sample collection and processing status so that progress is recorded.	The status update is saved against the correct appointment and is visible to authorized users.	High	Sprint-2
Lab Technician	Test Result Entry	USN-7	As a lab technician, I can enter and verify diagnostic test results so that reports can be prepared.	Result fields are saved and linked to the correct patient and test appointment.	High	Sprint-2
Patient	View Test Reports	USN-8	As a patient, I can view or download my completed diagnostic reports so that I can access my results.	A report is available only after it is finalized and only to authorized users for that patient.	High	Sprint-3
Administrator	Diagnostic Test Catalog Administration	USN-9	As an administrator, I can add, edit, or deactivate diagnostic test records so that the catalog remains current.	Authorized administrators can manage test records; inactive tests are not offered for new bookings.	High	Sprint-1
Administrator	Dashboard and User Management	USN-10	As an administrator, I can monitor bookings and manage authorized users so that system operations can be maintained.	Dashboard figures reflect stored records, and role-based access is enforced.	Medium	Sprint-3

3. Scope Notes

The scope of the Medical Diagnostic Center Management System is to manage patient registration, diagnostic test booking, appointment scheduling, sample processing, and report generation through the ServiceNow Service Portal. It allows patients to view tests and access reports, while lab technicians manage test results and administrators manage records, bookings, and diagnostic services.